

Multiple Choice (10 marks)

1. Use Laplace's equation to find the capacitance of a parallel plate capacitor.
 - (a) $C = (\epsilon^2 S / d)$
 - (b) $C = (/ d^2)$
 - (c) $C = (/ d)$
 - (d) $C = (\epsilon / dS)$
2. Speed of data transmission in 4-G network of telecom is
 - (a) 10 gbps - 100 gbps.
 - (b) 50 mbps - 500 mbps.
 - (c) 2 mbps - 1 gbps.
 - (d) 100 mbps - 1 gbps.
 - (e) 386 kbps - 2 mbps.
3. If a Hexadecimal number needs to convert to binary. For each hexadecimal digit, there will be how many bits
 - (a) 1
 - (b) 2
 - (c) 4
 - (d) 8
4. What is the name of the fluorescent material that gives red colour fluorescence?
 - (a) Zinc silicate.
 - (b) Calcium sulphide.
 - (c) Zinc oxide.
 - (d) Zinc sulphide.
 - (e) Calcium silicate.
5. The feedback factor of a Wien bridge oscillator using Op-Amp is
 - (a) $3/2$
 - (b) $1/4$
 - (c) 0
 - (d) $1/5$
 - (e) $1/3$

True / False (10 marks)

Answer True or False

6. True or False: The probability of obtaining 2 heads and 1 tail when tossing a penny 3 times is $3/4$
7. True or False: The correct answer to 'Find the input impedance of a 50 ohm line terminated in $+j$
8. True or False: The Laplace inverse of $F(s) = [(5s^2 - 15s + 7) / \{(s + 1)(s - 2)^3\}]$ is $e^{-t} + 2e^{2t}$
9. True or False: The normal acceleration for the fluid particle is given by $a_n = [(2ca) / \sqrt{(a^2 + b^2)}]$.
10. True or False: The probability of drawing one red ball from the box is $5/14$ and the probability of c

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the unit vector direction in which a positive test charge should be moved from the point (1, 3, -1) in the electric field $E = -xa_x + ya_y + 3a_z$ to achieve both maximum opposing and aiding forces.
12. Calculate the efficiency of a 7-hp motor that consumes 6.3 kilowatts at full load. Discuss the implications of this efficiency in practical engineering applications.

End of Paper